# Starstuff: getting the Seestar connection key

For testers of Starstuff 0.7.0-beta. About 15 minutes. A Windows PC and your own Android phone with the Seestar app.

## Why you need this

Since Seestar firmware 7.18, the telescope accepts commands only from software that proves itself with a key. The official Seestar phone app carries that key inside it. Starstuff's telescope add-on needs a copy of the key from *your own* copy of the Seestar app for Android. Without it, Starstuff can find the scope but every command is refused, and the Seestar tab shows "No RSA key is configured".

The key is copied to one file on your PC and never leaves it. Starstuff does not upload it, and the script below does not connect to anything.

**Please read this first**

- Starstuff contains no key and never downloads the Seestar app or the key.
- You read the key yourself, from a copy of the Seestar app you are entitled to use, on your own phone, solely so your own telescope accepts your own commands (interoperability).
- The key belongs to the maker of the Seestar app. Do not share it, post it or send it to anyone, including to us.
- Seestar and ZWO are trademarks of their owner. Starstuff is an independent project and is not affiliated with or endorsed by them.
- The script reads only the file you name, writes one file in your user folder and sends nothing. It is provided as is, without warranty.

**What you need**

- Starstuff installed, with the telescope add-on downloaded (Seestar tab, "Download Seestar support").
- The current Seestar app for **Android**, exported from your own phone as a package file: a `.apk` or `.xapk` file, usually 150 to 400 MB. The iPhone app does not work for this.
- The small script `Get-SeestarKey.ps1` that came with this guide.

## Step 1: get the Seestar app package file

Export the app from your own Android phone, the one you already use with the Seestar:

1. On the phone, install an "APK extractor" app from the Play Store (search for *APK extractor*; several free ones exist and any will do).
2. Open it, find **Seestar** in its list and choose Extract or Export. If it offers a choice, pick the bundle form ( `.xapk` or `.apks` ) so that all parts of the app are in one file.
3. Copy the file to the PC: USB cable, a cloud folder, or send it to yourself.

Use the current Seestar app. The file is your own copy of the app, as published by its maker. You are only reading the key out of a copy you are entitled to use.

## Step 2: run the script

1. Put `Get-SeestarKey.ps1` and the package file in the same folder, for example your Downloads folder.
2. Open that folder in File Explorer, right-click an empty spot and choose **Open in Terminal** (on older Windows: hold Shift, right-click, **Open PowerShell window here**).
3. Type this line and press Enter. Replace `Seestar.apk` with the real name of your file (for an `.xapk`, use that name):

```
powershell -ExecutionPolicy Bypass -File .\Get-SeestarKey.ps1 ".\Seestar.apk"
```

After a few seconds it prints:

```
Done. The key is saved at <your user folder>\.seestarpy\seestar.pem
Now start Starstuff (or restart it if it is running) and open the Seestar tab.
```

That is the whole job. The folder `.seestarpy` in your user folder now holds one small text file, `seestar.pem`.

## Step 3: check it in Starstuff

1. Start Starstuff. If it was already running, close its console window and start it again.
2. Open the **Seestar** tab. The "No RSA key is configured" warning is gone.
3. Put the Seestar in **Station mode** (see the box below), then press **Find scope** and connect.
4. The Firmware panel on the Seestar tab shows the key's fingerprint, a short code. That is how you know Starstuff is using it.

**The Seestar must be in Station mode**

Out of the box the Seestar runs its own Wi-Fi and your phone joins it. A PC cannot reach it that way. In the Seestar phone app, open the scope's settings, turn on **Station mode** and choose your home Wi-Fi. Put the PC on the same network. Then press Find scope in Starstuff.

## If something goes wrong

| What you see | What it means and what to do |
|---|---|
| "running scripts is disabled on this system" | Use the exact command line above: the `-ExecutionPolicy Bypass` part allows this one script to run. It changes nothing else on your PC. |
| "No key found in this package" | The file is not the Android Seestar app, or it is only one part of it. Export the bundle form ( `.xapk` or `.apks` ) from your own phone again. |
| "No such file" | The file name is mistyped, or the script and the package are not in the same folder. Keep the quotes around the name if it has spaces. |
| Starstuff still says "No RSA key is configured" | Starstuff reads the key when it starts. Close its console window and start it again. Then check that `%USERPROFILE%\.seestarpy\seestar.pem` exists. |
| Starstuff connects, but commands are refused | The key came from a very old or a modified app. Export the app from your phone again, after updating it to the current Seestar app. |
| You want the key somewhere else | Set the environment variable `SEESTAR_KEY_PATH` to the file's full path before starting Starstuff. |

Keep `seestar.pem` to yourself: do not post it or send it to anyone, including to us. It belongs to the maker of the Seestar app and is not part of Starstuff. If a Seestar firmware update changes the key, Starstuff will say so on the Seestar tab, and you repeat these steps with the then-current app.

## What the script does, for the curious

The app package is a zip file. Inside it, the file `lib/arm64-v8a/libopenssllib.so` holds the key as plain text between `-----BEGIN PRIVATE KEY-----` and `-----END PRIVATE KEY-----`. The script opens the zip (and any `.apk` inside an `.xapk`), finds that text and writes it to `seestar.pem`. It is about 60 lines of PowerShell and you can read it in Notepad. It uses nothing but what Windows already has; no Python, no extra tools.